

Small-Angle Scattering of Interacting Particles. I. Basic Principles of a Global Evaluation Technique

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Abstract

Indirect Fourier transformation is a well established method for the evaluation of small-angle scattering data. This technique, however, is restricted to dilute solutions, as for higher concentrations particle interactions can no longer be neglected. As the scattering intensity contains intra- and interparticle scattering contributions, the evaluation of scattering data is no longer just the solution of a linear weighted least-squares problem because the scattering intensity can, under certain conditions, be written as the product of the particle form factor and the so-called structure factor, which leads to a highly non-linear problem. In this paper a global evaluation technique including the structure factor is presented so that it is possible to determine the form factor and the structure factor simultaneously. This technique can be understood as a generalized version of the indirect Fourier transformation method. Like in the indirect Fourier transformation, there are no models or no analytical restrictions used for the form factor, and the structure factor is parameterized with up to four parameters for a given interaction model. A simultaneous determination of these two functions is possible due to the different analytical behavior of these functions, which also leads in most cases to the existence of a global minimum in the parameter surface. An algorithm to solve this non-linear least-squares problem has been developed and applied to simulated data for a variety of different uncharged systems.

1. Introduction

Small-angle scattering of X-rays and neutrons is a widely used technique to determine size, shape and internal structure of colloidal particles in the size range from a few nanometers up to about a hundred nanometers (Guinier & Fournet, 1955; Glatter & Kratky, 1982; Feigin & Svergun, 1987; Brumberger, 1995). In the case of very dilute solutions there exists a direct relation between the particle (represented by its pair distance distribution function) and its scattering data by means of a Fourier transformation. If concentrations

increase, however, the assumption of the average distance between the particles to be essentially larger than their dimensions is no longer valid. Thus, in addition to intraparticle scattering, interparticle scattering must be taken into consideration (Ailawadi, 1980; Hayter & Penfold, 1981; Hayter, 1985; Kaler, 1988, 1995; Hansen & McDonald, 1990; D'Aguanno & Klein, 1996). The former can be expressed by the so-called form factor, $P(q)$, where q is the length of the scattering vector. The latter corresponds to the structure factor, $S(q)$, that is necessary for the description of interacting systems and depends for spheres only on the relative locations of the individual particles.

Under certain conditions, the total scattering intensity, $I(q)$, can be represented or approximated by the product of the form factor $P(q)$ and the structure factor $S(q)$. The influence of $S(q)$ can easily be seen in experimental data: At low q values the scattering curve deviates from the ideal particle scattering curve. Repulsive interaction leads to a decrease of scattering intensity at small q values followed by a more or less pronounced interaction peak. These deviations at low q values affect the evaluation of experimental data so that strong oscillations in the pair distance distribution function, $p(r)$, may appear.

There exist various methods to remove the influence of interparticle contributions. In general it is possible to perform experiments with more than just one concentration and to extrapolate these scattering data to concentration zero. If the particle structure depends on the concentration, as it sometimes does, e.g. for aggregates of amphiphilic molecules, however, this method cannot be applied. Another possibility is to eliminate the inner part of the scattering curve, but this may lead to false results as the information on the size of elongated particles can be found in this q region. In quite a lot of cases one of the two factors is known from other experiments. If, for instance, a sample contains a monodisperse population of spheres, the structure factor $S(q)$ can be determined by a division of $I(q)$ by the known form factor $P(q)$. There is an interesting paper by Pedersen (1994) who tried to determine the number-density size distribution, $N(R)$, from small-angle scat-

tering data including the interparticle effects. His theory – the so-called local monodisperse approximation – is based on the assumption of a complete correlation of position and size of the particles, which is the case for a few special systems (e.g. precipitates in materials near dislocations in Al–Li alloys and gas bubbles near dislocations and grain boundaries in metals) (Mahalingam, Gu, Liedl & Sanders, 1987; Li, Kesternich, Schroeder, Schwahn & Ullmaier, 1990). In general, however, this assumption is not fulfilled, so that in contrast to his method our aim is to calculate $P(q)$ from small-angle scattering data at any concentration with a strict separation of intra- and interparticle contributions.

A new method which allows the simultaneous determination of the form factor and the structure factor with a minimum of *a priori* information is presented in this work. On the one hand, rough information about the maximum dimension of the particles is needed to determine $P(q)$. On the other hand, starting values for the volume fraction, radius, polydispersity and effective charge are necessary for the determination of $S(q)$.

As already mentioned, the scattering intensity $I(q)$ is made up of two terms, namely the form factor $P(q)$ and the structure factor $S(q)$. In the case of a monodisperse population of globular particles, the intraparticle scattering term is simply the square of the scattering amplitude $F(q)$. For a homogeneous and isotropic dispersion of spherical particles, no orientation effects must be taken into consideration so that $I(q)$ can be written as a product:

$$I(q) \propto NP(q)S(q). \quad (1)$$

Such a formal factorization is also possible for non-spherical systems. In these cases the structure factor has to be replaced by a so-called effective structure factor (for details see Weyerich, Brunner-Popela & Glatter, 1997). The indirect Fourier transformation (Glatter, 1977a,b, 1980a,b) allows the determination of the pair distance distribution function, $p(r)$, from $P(q)$ by means of the solution of a linear weighted least-squares problem including regularization. As can be seen in (1), $I(q)$ depends not only on $P(q)$ but also on $S(q)$, which gives rise to a highly nonlinear system if the two functions have to be determined simultaneously. In the following section we present a method to solve this nonlinear *ill-posed* problem. The aim is the simultaneous determination of $P(q)$ and $S(q)$, so we call the method a *Global Evaluation Technique* (program GET). It is based on the well established method of *Indirect Fourier Transformation for the determination of the Pair distance distribution function* (program ITP).

2. Theory

The method used is a generalization of the indirect Fourier transformation method, so we firstly give a brief

introduction to this technique. Starting in real space the pair distance distribution function $p(r)$ is approximated by a linear combination of a finite number of functions $\varphi_v(r)$

$$p(r) = \sum_{v=1}^n c_v \varphi_v(r), \quad (2)$$

where the basic functions $\varphi_v(r)$ are cubic B -splines that are defined in the subspace $0 \leq r \leq D_{\max}$ of the whole r space. The expansion coefficients c_v are the unknowns.

The above equations can be used for monodisperse and polydisperse systems. For polydisperse systems, one can alternatively calculate the size distribution $D(R)$. In this case, one has to assume a certain particle shape and the $p(r)$ function is simply replaced in (2) by the size distribution $D(R)$, and the $\sin(x)/x$ term in (4) has to be replaced by the form factor according to the shape assumed. In the following we only use the above notation for the calculation of $p(r)$, even though both possibilities are incorporated in our software.

$P(q)$ is the Fourier transform (T_1) of $p(r)$. Taking into account the linearity of this transformation, we can transform the particular functions φ_v into reciprocal space and obtain

$$\begin{aligned} P(q) &= T_1 p(r) \\ &= T_1 \sum_{v=1}^n c_v \varphi_v(r) \\ &= \sum_{v=1}^n c_v \psi_v(q), \end{aligned} \quad (3)$$

where

$$\begin{aligned} \psi_v(q) &= T_1 \varphi_v(r) \\ &= 4\pi \int_0^{D_{\max}} \varphi_v(r) (\sin qr / qr) dr. \end{aligned} \quad (4)$$

The ψ_v can be regarded as the basic functions for the form factor $P(q)$. Now the point is reached to include the structure factor according to (1):

$$I(q) = P(q)S(q, \mathbf{d}) = \sum_{v=1}^n [c_v \psi_v(q)] S(q, \mathbf{d}). \quad (5)$$

The vector $\mathbf{d}$ contains the coefficients d_k ($k = 1-4$) determining the structure factor. These coefficients correspond to the volume fraction, the size (radius), the polydispersity parameter and the charge of the particle. As can be seen from (5), the scattering intensity $I(q)$ is no longer just a function of the scattering vector q and the coefficients c_v but from that point on also of the coefficients d_k . The scattering function can thus be represented as the product of the Fourier-transformed B -splines $\psi_v(q)$ and the structure factor.

The functions $\psi_v(q)$ can, for simplicity, be directly multiplied by $S(q, \mathbf{d})$ defining a new set of functions $\tilde{\psi}_v$

for the representation of the scattering intensity $I(q)$ from interacting systems:

$$\overline{\psi}_v(q, \mathbf{d}) = \psi_v(q)S(q, \mathbf{d}). \quad (6)$$

With that definition we obtain

$$I(q) = \sum_{v=1}^n c_v \psi_v(q)S(q, \mathbf{d}) = \sum_{v=1}^n c_v \overline{\psi}_v(q, \mathbf{d}). \quad (7)$$

Thus the scattering function $I(q)$ is described as a linear combination of the functions $\overline{\psi}_v$ that depend both on q and on $\mathbf{d}$ and therefore implicitly contain information about the structure factor.

The smearing effects with regard to wavelength distribution (formally represented by transformation T_2) which are very important in the case of small-angle neutron scattering and geometric effects (transformations T_3 and T_4) can also be taken into account (for further detail see Glatter, 1977b):

$$\tilde{I}(q) = T_4 T_3 T_2 I(q) = \sum_{v=1}^n c_v \overline{\chi}_v(q, \mathbf{d}) \quad (8)$$

$$\overline{\chi}_v(q, \mathbf{d}) = T_4 T_3 T_2 \overline{\psi}_v(q, \mathbf{d}), \quad (9)$$

where $\tilde{I}(q)$ represents the smeared scattering intensity which is described by the basic functions $\overline{\chi}_v$.

The coefficients c_v and d_k can be determined by a weighted least-squares approximation of the experimental data $I_{\text{exp}}(q)$ by the sum over all basic functions (Bevington, 1969; Brandt, 1970)

$$L = \sum_{i=1}^m \{I_{\text{exp}}(q_i) - \tilde{I}(q_i, \mathbf{c}, \mathbf{d})\}^2 / \sigma(q_i)^2 \\ = \sum_{i=1}^m \{I_{\text{exp}}(q_i) - \sum_{v=1}^n c_v \overline{\chi}_v(q_i, \mathbf{d})\}^2 / \sigma(q_i)^2 = \min, \quad (10)$$

where m is the number of data points, $\sigma^2(q_i)$ is the variance of the i th point, the vector $\mathbf{c}$ contains the coefficients c_v defining $P(q)$, and the vector $\mathbf{d}$ contains the parameters of the structure factor $S(q)$.

The main difficulty, however, is that in our case the least-squares problem leads to a highly nonlinear system of equations because of the product of form factor and structure factor. In principle, there exist two possibilities to solve this problem.

On the one hand, one could perform directly a nonlinear weighted least-squares procedure with regard to all parameters c_v and d_k . Thus, in the extreme there should be found a global minimum in a $(n+4)$ -dimensional (!) hypersurface where the number n can be as high as 40 (corresponding to the n B -splines describing the pair distance distribution function) and four parameters d_k [corresponding to $S(q)$]. Furthermore, as is shown later in more detail, there arise stability problems on solution of the least-squares approximation with regard to the coefficients c_v , so that for such an ill-posed problem a stabilization constraint must be taken into

account. If both c_v and d_k are determined within one procedure there should be found a type of partially effective stabilization, which we found to be quite complicated.

On the other hand, one could follow a new path by splitting the problem into two parts referring to the different nature of the two parameter sets c_v and d_k . Therefore, there is one procedure to solve a linear weighted least-squares problem to determine the parameters c_v for the form factor for a – for the moment – constant set of coefficients d_k , and there is a second procedure, a nonlinear weighted least-squares routine, for the determination of the parameters d_k for the structure factor, while the c_v parameters are kept constant in this step. These two procedures are highly correlated, as is shown below.

L is minimized when the partial derivatives with respect to c_v and d_k vanish simultaneously, i.e. if

$$\frac{\partial L}{\partial c_v} = 0 \quad \text{and} \quad \frac{\partial L}{\partial d_k} = 0. \quad (11)$$

Finally, under consideration of (8) these requirements lead to the partial derivatives of $\tilde{I}(q)$ with respect to c_v and d_k :

$$\frac{\partial \tilde{I}(q)}{\partial c_v} = \overline{\chi}_v(q, \mathbf{d}) \quad v = 1, 2, \dots, n, \quad (12)$$

$$\frac{\partial \tilde{I}(q)}{\partial d_k} = \sum_{v=1}^n c_v \frac{\partial \overline{\chi}_v(q, \mathbf{d})}{\partial d_k} \quad k = 1, 2, \dots, 4. \quad (13)$$

As described above, there are two types of parameter sets that are calculated separately. The first step is to keep $\mathbf{d}$ fixed as $\mathbf{d}^{(0)}$ in order to solve the variational problem in (10) with regard to $\mathbf{c}$ by means of normal equations. Then, the solution coefficients $\mathbf{c}$ can be used to determine improved values for $\mathbf{d}^{(1)}$ by another type of normal equations. Later, $\mathbf{c}$ is calculated again with the new and fixed $\mathbf{d}^{(1)}$ etc. Thus, the two coefficients are optimized separately but simultaneously in an iterative procedure.

2.1. Determination of the form-factor coefficients $\mathbf{c}$

Solving (10) for $\mathbf{c}$ (containing the coefficients c_v) with fixed $\mathbf{d}$ leads to the normal equations

$$\mathbf{b} = \mathbf{B}\mathbf{c}, \quad (14)$$

where the elements of the vector $\mathbf{b}$ are given by

$$(\mathbf{b})_\mu = b_\mu = \sum_{i=1}^m \overline{\chi}_\mu(q_i, \mathbf{d}) [1/\sigma^2(q_i)] I_{\text{exp}}(q_i) \quad (15)$$

and the elements of the matrix $\mathbf{B}$ by

$$(\mathbf{B})_{\mu\nu} = B_{\mu\nu} = \sum_{i=1}^m \overline{\chi}_\mu(q_i, \mathbf{d}) [1/\sigma^2(q_i)] \overline{\chi}_\nu(q_i, \mathbf{d}). \quad (16)$$

As stability problems usually arise for such systems, (14) has to be modified by a regularization term $\lambda \mathbf{K}$:

$$\mathbf{b} = (\mathbf{B} + \lambda \mathbf{K})\mathbf{c}, \quad (17)$$

where the matrix $\mathbf{K}$ is defined as

$$\mathbf{K} = \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & & \ddots & & \\ & -1 & & 2 & -1 \\ & & & -1 & 1 \end{pmatrix} \quad (18)$$

and λ is a free stabilizing parameter (Glatter, 1977a,b). These modified normal equations are solved by inversion of the matrix $(\mathbf{B} + \lambda \mathbf{K})$,

$$\mathbf{c} = (\mathbf{B} + \lambda \mathbf{K})^{-1} \mathbf{b}, \quad (19)$$

or by direct methods (Press, Teukolsky, Vetterling & Flannery, 1992).

2.2. Determination of the structure-factor coefficients $\mathbf{d}$

After the calculation of the parameters $\mathbf{c}$, these parameters can be used to determine improved values for $\mathbf{d}$ with fixed $\mathbf{c}$. Using (6) and (9), the partial derivatives in (13) are given by

$$\begin{aligned} \frac{\partial \bar{\chi}_v}{\partial d_k} &= \frac{\partial}{\partial d_k} T_4 T_3 T_2 \psi_v(q) S(q, \mathbf{d}) \\ &= T_4 T_3 T_2 \psi_v(q) \frac{\partial S(q, \mathbf{d})}{\partial d_k}. \end{aligned} \quad (20)$$

As the structure factor is the only factor containing $\mathbf{d}$, the partial derivatives of $S(q, \mathbf{d})$ with respect to these parameters must be calculated:

$$S^k = \frac{\partial S(q, \mathbf{d})}{\partial d_k}. \quad (21)$$

According to the definitions of $\bar{\psi}_v$ in (6) and $\bar{\chi}_v$ in (9), the product of S^k and $\psi_v(q)$ can be defined as $\bar{\psi}_v^k$ and after performance of the smearing effects as $\bar{\chi}_v^k$:

$$\bar{\psi}_v^k(q, \mathbf{d}) = \psi_v(q) S^k(q, \mathbf{d}), \quad (22)$$

$$\bar{\chi}_v^k(q, \mathbf{d}) = T_4 T_3 T_2 \bar{\psi}_v^k(q, \mathbf{d}). \quad (23)$$

Using these definitions the partial derivatives in (13) lead to a new set of functions:

$$\tau_k(q, \mathbf{c}, \mathbf{d}) = \frac{\partial \tilde{I}(q)}{\partial d_k} = \sum_{v=1}^n c_v \bar{\chi}_v^k(q, \mathbf{d}) \quad k = 1, 2, \dots, 4. \quad (24)$$

Thus, the derivatives of $\tilde{I}(q)$ with respect to d_k in (13) are the functions τ_k which are the sum over all n functions $\bar{\chi}_v^k$ weighted by the coefficients c_v .

As the function $\tilde{I}(q)$ is nonlinear in its coefficients $\mathbf{d}$, the solution of the least-squares problem is no longer as

easy as described before. The nonlinear weighted least-squares problem can just be solved iteratively by means of a Taylor expansion:

$$\begin{aligned} I_{\text{exp}}(q) &= \tilde{I}(q, \mathbf{d}^{(n)}) + \sum_{k=1}^4 \frac{\partial \tilde{I}(q, \mathbf{d}^{(n)})}{\partial d_k} (d_k^{(n+1)} - d_k^{(n)}) \\ &\quad + \text{high-order terms.} \end{aligned} \quad (25)$$

All high-order terms are neglected so that (25) can be rewritten as

$$I_{\text{exp}}(q) - \tilde{I}(q, \mathbf{d}^{(n)}) = \sum_{k=1}^4 \frac{\partial \tilde{I}(q, \mathbf{d}^{(n)})}{\partial d_k} (d_k^{(n+1)} - d_k^{(n)}). \quad (26)$$

The error introduced by neglect of the higher order terms necessitates an iterative procedure for the final solution of the problem. The partial derivatives of $\tilde{I}(q)$ with respect to d_k have already been defined as τ_k [(24)], which leads to

$$\Delta I(q, \mathbf{d}^{(n)}) = \sum_{k=1}^4 \tau_k^{(n)}(q, \mathbf{c}, \mathbf{d}) \Delta d_k^{(n)} \quad (27)$$

for all values of q , where

$$\Delta I(q, \mathbf{d}^{(n)}) = I_{\text{exp}}(q) - \tilde{I}(q, \mathbf{d}^{(n)}) \quad (28)$$

and

$$\Delta d_k^{(n)} = d_k^{(n+1)} - d_k^{(n)} \quad \text{or} \quad \Delta \mathbf{d}^{(n)} = \mathbf{d}^{(n+1)} - \mathbf{d}^{(n)} \quad (29)$$

in vector notation.

The least-squares solution of (27) for the determination of the coefficients $\mathbf{d}$ leads to a second set of normal equations

$$\mathbf{t} = \mathbf{T} \Delta \mathbf{d}^{(n)}, \quad (30)$$

where, according to (15) and (16), the vector $\mathbf{t}$ and the matrix $\mathbf{T}$ are defined as

$$(\mathbf{t})_\mu = t_\mu = \sum_{i=1}^m \tau_\mu(q_i, \mathbf{c}, \mathbf{d}) [1/\sigma^2(q_i)] \Delta I(q_i) \quad (31)$$

and

$$(\mathbf{T})_{\mu\nu} = T_{\mu\nu} = \sum_{i=1}^m \tau_\mu(q_i, \mathbf{c}, \mathbf{d}) [1/\sigma^2(q_i)] \tau_\nu(q_i, \mathbf{c}, \mathbf{d}). \quad (32)$$

The solution coefficients $\Delta \mathbf{d}^{(n)}$ can easily be calculated by inversion of matrix $\mathbf{T}$:

$$\Delta \mathbf{d}^{(n)} = \mathbf{T}^{-1} \mathbf{t}. \quad (33)$$

With (29) we can now calculate the improved coefficients $\mathbf{d}^{(n+1)}$ which are the starting point for the next iteration. This iteration can be called the $\mathbf{d}$ loop.

2.3. Combination of linear and nonlinear weighted least-squares systems

In summary, there are two main parts in the simultaneous determination of the form factor and of the structure factor. On the one hand, the parameters $\mathbf{c}$ must be calculated for an assumed vector $\mathbf{d}$ to determine the form factor. The stability problem is solved by a so-called λ loop varying the Lagrange multiplier to find an optimum solution for (17) for the fixed vector $\mathbf{d}$. On the other hand, the parameters $\mathbf{d}$ determining the structure factor have to be improved with fixed $\mathbf{c}$ by an iterative algorithm to solve the nonlinear weighted least-squares system ($\mathbf{d}$ loop). With this improved vector $\mathbf{d}$, we jump back to the calculation of $\mathbf{c}$ in the λ loop and so on until a final convergence is reached.

2.4. Structure factor

In principle, four parameters are reserved for the description of the structure factor $S(q)$, namely the volume fraction ϕ , the size (radius) R_{HS} , the polydispersity parameter μ and the charge of the particles z . In the literature there can be found quite a number of different structure factors dependent on the theory used. As a first step, we have chosen an analytical solution of $S(q)$ for hard spheres without any charge because the basic features already exist in the case of uncharged systems. We therefore refer to the analytical expressions in the Percus–Yevick (PY) approximation (Ornstein & Zernike, 1914; Percus & Yevick, 1958; Percus, 1962; Wertheim, 1963; Ashcroft & Lekner, 1966; Kinning & Thomas, 1984; Pedersen, 1994).

The corresponding analytical formula contains the volume fraction and the radius of the hard spheres as parameters without any kind of polydispersity, which leads to very pronounced oscillations in $S(q)$. From practical experience we know, however, that such a structure factor is hardly ever observed owing to the polydispersity of the systems studied (Strey, 1992, 1994; Gradzielsky, Langevin, Magid & Strey, 1995). In order to damp these oscillations, we introduce a polydispersity parameter by averaging different structure factors for a certain size distribution to obtain an ‘averaged’ structure factor $S^{av}(q)$. Each of these $S_\alpha(q)$ functions is calculated for a monodisperse system of spheres (species α , molar fraction x_α) of varying radii but with fixed volume fraction ϕ :

$$S^{av}(q) = \sum_{\alpha} x_{\alpha} S_{\alpha}(q). \quad (34)$$

We define μ as the ratio between the width of the Gaussian distribution (σ) and its maximum value (R_{HS}), i.e. $\mu = \sigma/R_{HS}$.

The influence of the various parameters on the structure factor is illustrated in Fig. 1. Fig. 1(a) shows the influence of the volume fraction ϕ . If ϕ increases, the oscillations become more and more pronounced and

the maxima are slightly shifted to higher q values; this means the higher the concentration, the higher the order of the particles and the closer the distances. The value of

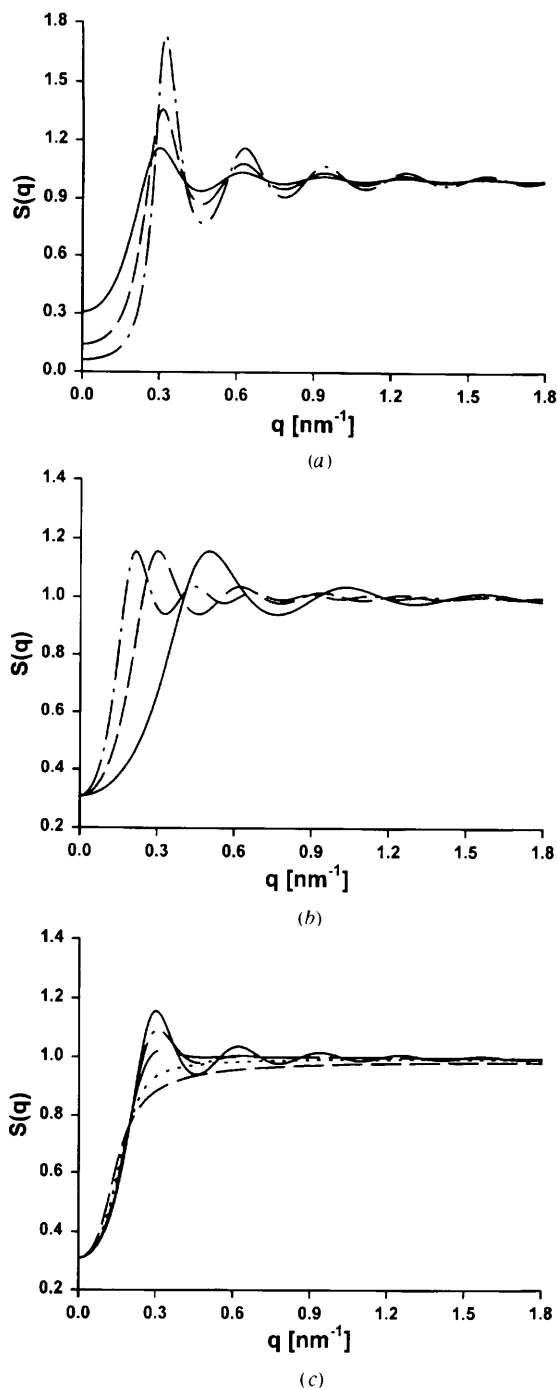


Fig. 1. (a) Structure factors for varying volume fraction ϕ : — $\phi = 0.15$, --- $\phi = 0.25$, - - - $\phi = 0.35$ and fixed $R_{HS} = 10$ nm, $\mu = 0.0$. (b) Structure factors for varying radii R_{HS} : — $R_{HS} = 6$ nm, --- $R_{HS} = 10$ nm, - - - $R_{HS} = 14$ nm and fixed $\phi = 0.15$, $\mu = 0.0$. (c) Structure factors for varying polydispersity μ : — $\mu = 0.0$, --- $\mu = 0.15$, - - - $\mu = 0.25$ nm, ···· $\mu = 0.4$, --- $\mu = 0.6$ and fixed $R_{HS} = 10$ nm, $\phi = 0.15$.

$S(0)$ changes significantly, which can be explained by the isothermal osmotic compressibility (the system is less compressible in the case of higher concentration). If the radius of the hard spheres, R_{HS} , is increased, the maxima are shifted to the lower q region corresponding to the increasing distance to neighbouring particles (Fig. 1b). In other words, these three functions would coincide in a plot of $S(q)$ versus qR_{HS} . Finally, the increase of the polydispersity parameter μ leads to a smoothing of the oscillations. The integration width for the polydispersity is proportional to q and therefore there is no influence on $S(0)$. The averaging effect is strongest in the regime of the first side maximum and becomes negligible again at very high q where $S(q)$ is flat and simply equal to unity (Fig. 1c).

Nevertheless, we know that this is not a correct representation of polydispersity effects as there is absolutely no influence on $S(0)$ (Vrij, 1979). In the case of monodisperse globular particles, $S(0)$ is related to the isothermal osmotic compressibility χ_T . For polydisperse systems, however, this relation does not hold any longer because there is a great influence of the polydispersity on $S(0)$ although χ_T is practically constant. Nevertheless, within this paper we restrict ourselves to this simplified description. A correct description for the polydispersity effect has been worked out in parallel to this work (Weyerich, Brunner-Popela & Glatter, 1997) and will be incorporated in an improved version of this technique.

3. Numerical results

The first step on the way to a global evaluation of small-angle scattering data was to examine whether the parameters of the structure factor can be optimized in parallel to the estimation of $P(q)$; this means we had to check if there exists a global minimum in the parameter surface. Thus we started a feasibility study without the algorithm described above by simply calculating the product of $P(q)$ and $S(q)$ for different combinations of the parameters of the structure factor $\mathbf{d}$ while determining $P(q)$ as described above.

Once the principal possibility of determination of the parameters of $S(q)$ was established, we had to create an algorithm to automate the search for a global minimum, which led to the procedure described above.

3.1. Global minimum

One of the most important points with regard to the simultaneous determination of $P(q)$ and $S(q)$ is whether there exists a global minimum at all. In order to check convergence problems with regard to the nonlinear least-squares system in advance, we started with simulations.

First we simulated the form factor $P(q)$ and the structure factor $S(q)$ for a given system. The product of

Table 1. Dependence of the mean deviation on the parameters ϕ and R_{HS} of $S(q)$, simulated for $\phi = 0.15$, $R_{HS} = 10$ nm, $\mu = 0.0$

ϕ/R_{HS}	7	8	9	10	11	12	13
0.075	13.44	11.03	8.53	6.60	7.43	10.02	13.23
0.100	15.13	11.77	7.92	4.72	6.53	10.94	15.59
0.125	17.14	12.92	7.20	2.71	6.55	12.79	18.93
0.150	19.35	14.36	7.51	0.97	7.79	15.71	22.84
0.175	21.68	15.42	8.34	2.71	10.46	19.22	27.40
0.200	23.09	17.16	9.57	5.68	13.72	23.43	33.28
0.225	25.32	18.97	11.28	8.96	17.70	28.29	39.06

these two terms led to simulated data points where resolution effects and statistical noise were added. From these data, solutions for c_v [corresponding to $P(q)$] were calculated with fixed d_k [corresponding to $S(q)$]. A systematic variation of the parameters d_k allowed us to check if there existed a global minimum. The mean deviation was used as a measure of the goodness of the fit.

In order to visualize the principal possibility, we show in the following a simplified example, where $S(q)$ depends only on two parameters, namely the volume fraction ϕ and the radius of hard spheres R_{HS} . The polydispersity is neglected in this case; a wavelength distribution (FWHM = 15%) as it often occurs in small-angle neutron scattering experiments and a normally distributed statistical error are taken into account. In Table 1 the values for the mean deviation (MD) are listed according to the $S(q)$ parameters, and these data are shown in a three-dimensional plot (Fig. 2), where for better visibility $(MD + 1)^{-1}$ is plotted. As can be seen there exists a clear optimum. We have found such optimum solutions (minimum in the mean deviation) for many different parameter sets. This is not yet a general proof that a global minimum exists in all cases but

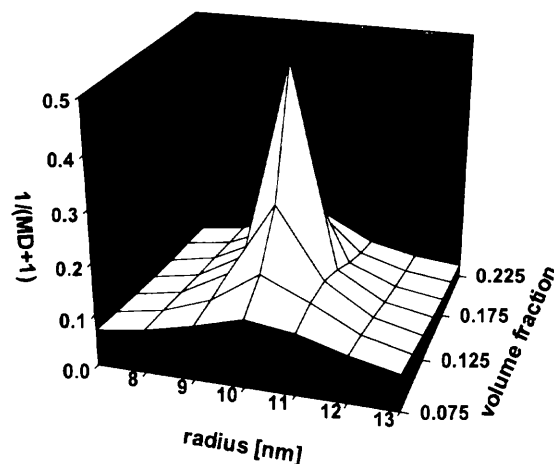


Fig. 2. Parameter optimization: the three-dimensional plot of the inverse mean deviation $(MD + 1)^{-1}$ versus the $S(q)$ parameters ϕ and R_{HS} shows a clear optimum.

justifies the development of an automatic procedure for the estimation of the optimum parameter set d_k .

3.2. Simulations

The next section shows simulations for different typical experimental situations including the limits and the accuracy of the method. The principle is always the same. We start with the simulation of a form factor and a structure factor (according to the PY approximation) that is described by the parameters ϕ (volume fraction), R_{HS} (radius of hard spheres) and μ (polydispersity parameter). In all these simulations the influence of charge is absolutely neglected. Smearing effects according to a typical small-angle neutron scattering experiment (wavelength distribution, FWHM = 15%), in the following denoted as SANS, or smearing effects according to a Kratky system (slit length effect, slit width effect, and 25% $K\beta$ contribution), in the following denoted as SAXS, are taken into account in our simulations. In addition, statistical noise is added. This simulated scattering intensity $I(q)$ is evaluated for the determination of the form factor, represented in real space by the pair distance distribution function $p(r)$, and the structure factor is determined simultaneously. Starting values for $S(q)$ are needed to optimize the parameters d_k by means of the algorithm described in the previous chapter.

We show the results for the following typical examples set out of a large series of simulations:

- (a) homogeneous systems:
 - (i) homogeneous spheres, point collimation and slit collimation;
 - (ii) homogeneous nonspherical particles (cylinders).
- (b) inhomogeneous systems:
 - (i) inhomogeneous spheres, point collimation and slit collimation;
 - (ii) inhomogeneous cylinders.

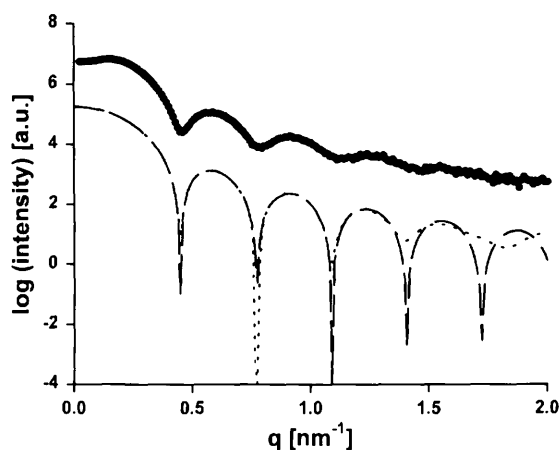


Fig. 3. Homogeneous spheres, scattering functions: simulation, — approximation, — theoretical form factor, - - - estimated form factor.

(c) polydisperse systems:

- (i) polydisperse distribution of homogeneous spheres, calculation of the volume distribution $D_v(R)$.

3.2.1. *Homogeneous systems.* In the first example, a homogeneous sphere, we have chosen the radius of the spheres to be 10 nm. The simulated structure factor has the following parameters: $\phi = 0.15$ (0.18), $R_{\text{HS}} = 10$ (12) nm, $\mu = 0.4$ (0.5). The values in parentheses are the starting values used for the solution of the problem. It turned out that the procedure always showed better convergence with overestimated starting values.

In Fig. 3, the simulated data points (SANS) are plotted as well as the approximation from the evaluation procedure. By comparison of the theoretical form factor with the resulting $P(q)$, a perfect agreement can be observed. The corresponding pair distance distribution functions $p(r)$ are shown in Fig. 4; both for SANS and SAXS simulation the result is in excellent agreement with the theoretical function.

Using the form factor for a homogeneous nonspherical particle, a cylindrical particle with the axis parameters 11 nm diameter and 21 nm length, and the structure factor (for hard-spheres interaction) with $\phi = 0.15$ (0.19), $R_{\text{HS}} = 12$ (16) nm, and $\mu = 0.4$ (0.6), can be evaluated as well. In Fig. 5, the theoretical $p(r)$ and the resulting $p(r)$ are plotted. The good agreement between these two functions shows that our new procedure can also be used for the evaluation of scattering data of nonspherical particles as long as orientational interaction can be neglected.

3.2.2. *Inhomogeneous systems.* The next step was to use our new method for the evaluation of monodisperse, inhomogeneous systems still using the structure factor for homogeneous spheres (PY approximation).

In Fig. 6, we show the scattering function using the form factor for an inhomogeneous sphere (core-shell model, ten-steps model, $\rho_i < 0$, $\rho_a > 0$, $R_i = 4$ nm, $R_a = 10$ nm; insert of Fig. 8). As usual, smearing effects

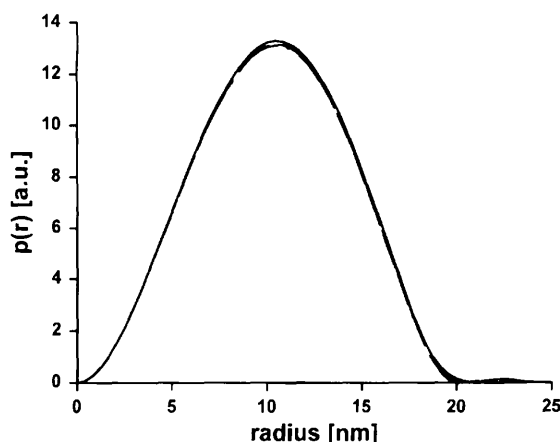


Fig. 4. Homogeneous spheres, pair distance distribution functions: — theoretical $p(r)$, — estimated $p(r)$ – SANS, - - - estimated $p(r)$ – SAXS.

according to SANS (point collimation) are taken into account, and a relatively large statistical noise is added (4% at low q values). The limiting q values are 0.05 and 1.19 nm^{-1} . The structure factor taken into account is calculated for the parameters $\phi = 0.285$ (0.32), $R_{\text{HS}} = 10$ (12) nm, $\mu = 0.3$ (0.4), and plotted in Fig. 7. The evaluation procedure enables the determination of both the form factor and the structure factor. The fit is perfect, which is visualized in Figs. 6 and 7; both $P(q)$ and $S(q)$ coincide excellently. The pair distance distribution functions are shown in Fig. 8. In the case of 4% error on the data points there are very small deviations in the outer part of $p(r)$. For 1% error the agreement with the theoretical function is again almost perfect.

It is even possible to use data simulated for a Kratky camera. The form factor corresponds to an inhomogeneous

sphere as described above, and the structure factor is calculated for the parameters $\phi = 0.15$ (0.18), $R_{\text{HS}} = 10$ (12) nm, $\mu = 0.4$ (0.5). Smearing effects according to slit collimation are taken into account as well as statistical noise (1% at low q values). The limiting q values are 0.1 and 1.19 nm^{-1} . Even in this case the structure factor can be determined from the simulated scattering data so that the agreement of the theoretical form factor and the resulting one is extremely good (Fig. 9). The $p(r)$ functions are plotted in Fig. 10 and confirm this trend.

These two examples prove that, even in the case of relatively poor scattering data (severe termination in q space, relatively large error bars, distinct smearing effects, and great influence of interparticle contributions), a simultaneous determination of $P(q)$ and $S(q)$ is possible.

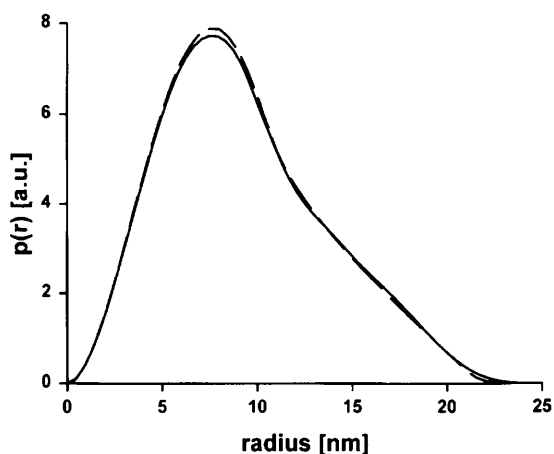


Fig. 5. Homogeneous cylinders, pair distance distribution functions: — theoretical $p(r)$, — estimated $p(r)$ - SANS.

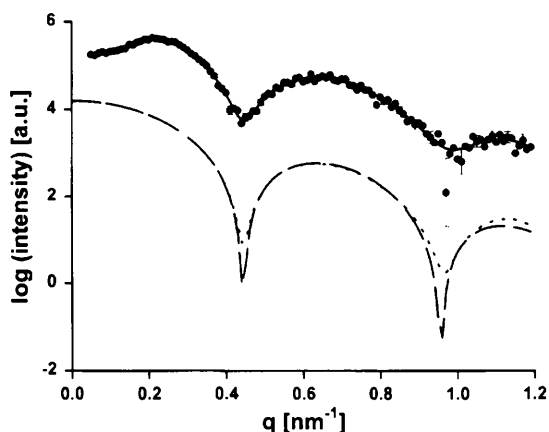


Fig. 6. Inhomogeneous spheres, scattering functions - SANS: simulation, — approximation, --- theoretical form factor, - · - · - estimated form factor.

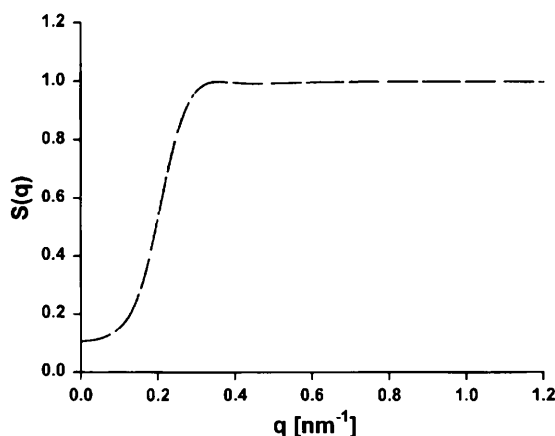


Fig. 7. Inhomogeneous spheres, structure factors - SANS: --- simulated structure factor, - · - · - estimated structure factor.

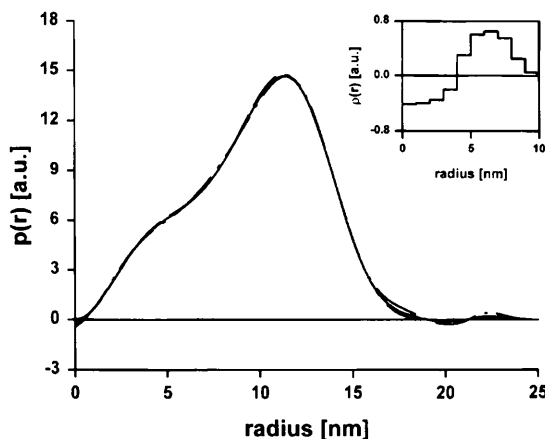


Fig. 8. Inhomogeneous spheres, pair distance distribution functions: --- theoretical $p(r)$, — estimated $p(r)$ - SANS (1% error on data points), - · - · - estimated $p(r)$ - SANS (4% error on data points). Insert: scattering-length density distribution used for the simulation.

Finally, we have tried to evaluate simulated scattering data of an inhomogeneous cylinder. The form factor is calculated for the following parameters: 11 nm diameter and 29 nm length, $\rho_i = -0.8$, $\rho_a = 1$, $R_i = 2.5$ nm, $R_a = 5.5$ nm. The simulated structure factor has the following parameters: $\phi = 0.15$ (0.18), $R_{HS} = 12$ (14) nm, $\mu = 0.4$ (0.5). As usual, smearing effects (SANS) and statistical noise are added. In Fig. 11 the theoretical and the estimated $p(r)$ functions are plotted, and the agreement is excellent again; the deviations from the theoretical curve are within the error bars.

3.2.3. Polydisperse systems. Finally, we want to show that small-angle scattering data of polydisperse systems can also be evaluated using our new method. Thus the form factor is calculated for a particle size distribution function $D_v(R)$ of homogeneous spheres [according to equation 1(b) in Glatter (1980a)] as shown in Fig. 13, a Gaussian distribution with $R_{max} = 12$ nm and $\sigma = 2$. The resolution effects according to SANS

and statistical noise are added. The structure factor is calculated for the parameters $\phi = 0.15$ (0.18), $R_{HS} = 12$ (15) nm, $\mu = 0.4$ (0.6). We should mention again that no real polydispersity for the structure factor is taken into account in this example. In Fig. 12 the theoretical and the estimated form factors are plotted as well as the approximation to the simulated scattering data. The fit is very good, which can also be seen in Fig. 13 with regard to the corresponding size distribution functions $D_v(R)$; the deviations are within the error bars.

4. Discussion

The local evaluation technique, introduced in this paper, allows the simultaneous determination of the form factor and the structure factor performing a combined linear/nonlinear weighted least-squares procedure with a

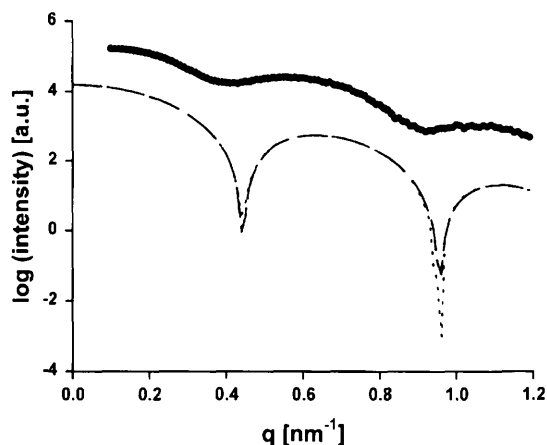


Fig. 9. Inhomogeneous spheres, scattering functions – SAXS: simulation, — approximation, --- theoretical form factor, -.-.- estimated form factor.

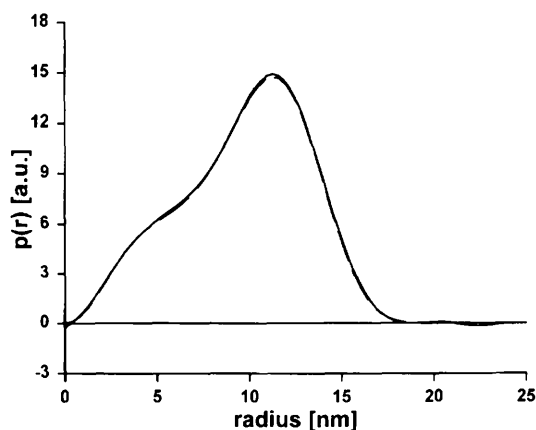


Fig. 10. Inhomogeneous spheres, pair distance distribution functions: --- theoretical $p(r)$, — estimated $p(r)$ – SAXS.

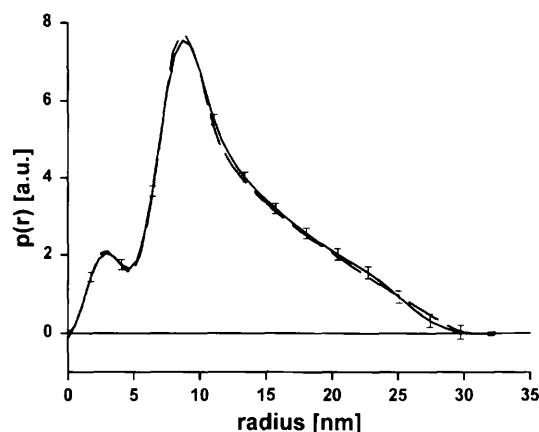


Fig. 11. Inhomogeneous cylinders, pair distance distribution functions: --- theoretical $p(r)$, — estimated $p(r)$ – SANS.

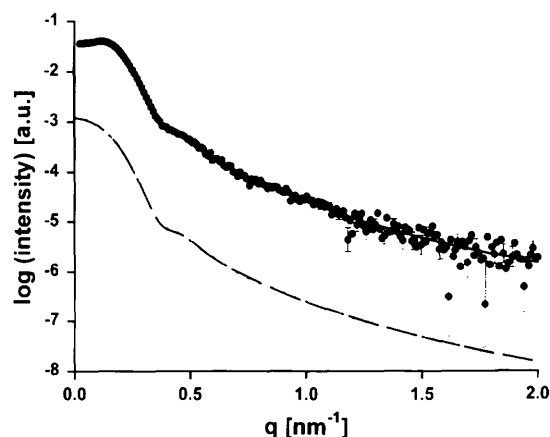


Fig. 12. Polydisperse spheres, scattering functions: simulation, — approximation, --- theoretical form factor, -.-.- estimated form factor.

minimum of *a priori* information. Our approach does not compare with usual model fitting as it is based on a free determination of the form factor without any assumption on particle shape, size or internal structure. We only assume a model for the representation of the structure factor that can in principle be any type of model; in this first paper we restrict ourselves to the hard-spheres interaction including polydispersity *via* an averaged structure factor $S^{\text{av}}(q)$. We have shown various examples to prove that in fact it is possible to find a global minimum in the parameter surface. Although these first results are very promising, some points need critical discussion and further work.

The first essential point is the factorization of the scattering intensity $I(q)$ into the product of the particle form factor $P(q)$ and the structure factor $S(q)$, which is strictly valid only for monodisperse spheres. Any violation of this basic constraint, such as polydispersity, nonspherical shape and orientation-dependent interaction, would not allow such a factorization where the structure factor is only a function of the particle positions.

It is however possible to define an *effective* or *measured* structure factor by dividing $I(q)$ by a mean particle form factor $\overline{P(q)}$. Such an effective factor $S^{\text{eff}}(q)$ would depend on the scattering amplitudes of the individual particles, and no general method exists to calculate such an effective structure factor. It can, however, be calculated for an ensemble of polydisperse spheres with hard-spheres interaction. It can be shown that this effective structure factor gives a perfect representation in this case and is therefore favorable compared with the simple averaged structure factor and with the decoupling approximation (Weyerich, Brunner-Popela & Glatter, 1997). This concept of an effective structure factor can, in principle, also be applied to nonspherical particles (Weyerich, D'Aguanno, Canessa & Klein, 1990).

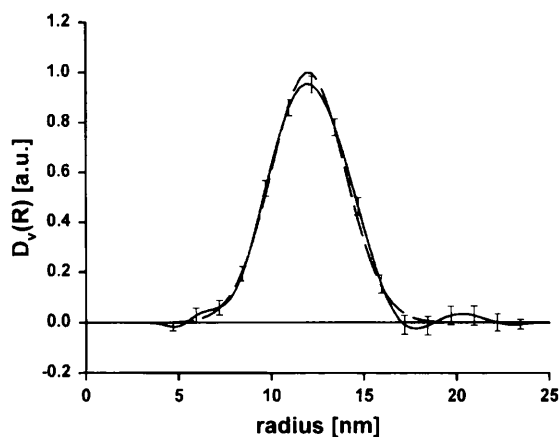


Fig. 13. Polydisperse spheres, volume distribution: ---- theoretical $D_v(R)$, — estimated $D_v(R)$ – SANS.

In our global evaluation method we are primarily interested in determining the particle scattering functions, *i.e.* the form factor $P(q)$ and its Fourier transform, the pair distance distribution function $p(r)$, and the structure factor is just used to take into account inter-particle scattering contributions. This means we keep the form factor as free as possible, and there are no assumptions on the particle shape, size and the size distribution or on the internal structure as it was already described in previous papers. With regard to inter-particle contributions, however, we have to choose a special type of structure factor, and in this paper it is $S(q)$ with a hard-spheres potential, which is, for example, quite a good description for microemulsions. Furthermore, to our knowledge there exists just a small number of solutions for the structure factor for polydisperse uncharged or charged systems, therefore the interaction potential cannot be chosen arbitrarily. In any case, our technique is not meant to be used to determine correct interaction models and their parameters. We only want to cover the interaction contributions as well as possible to obtain reliable information on the form factor $P(q)$.

Now we want to discuss the assumption on $P(q)$ and $S(q)$ in more detail. The particle scattering functions should be as arbitrary as possible (arbitrary shapes and inhomogeneities should be possible), and the $p(r)$ function should be estimated with minimized termination effect or cut-off effect due to the limited q range of experimental data. The concept of indirect Fourier transformation has proven to fulfil all these conditions under the assumption of a maximum particle size using a limited number of cubic B -splines to model the real-space function. It also allows for the correction of instrumental broadening (*desmearing*). In the case of polydisperse systems a particle shape has to be assumed and the corresponding size distribution $D(R)$ results instead of the $p(r)$ function.

The parameterization of the structure factor is more difficult. This technique is not at all designed to extract reliable physical information from the structure factor like interaction potentials. We assume a certain interaction – in this contribution only hard-spheres interaction – and model the structure factor with a few parameters such as volume fraction ϕ , hard-sphere radius R_{HS} , and averaging according to a Gaussian size distribution with a variable relative width μ . We have to assume that the measured structure factor can be approximated under these constraints sufficiently well. We know, for instance, that these assumptions will not be correct for very dense systems and may not be fulfilled for polydisperse or highly nonspherical objects, but it is at least a step in the right direction. Using an averaged hard-spheres structure factor for elongated particles such as prolate ellipsoids or cylinders can be justified as long as the orientational interaction is neg-

ligible, *i.e.* when the concentration is not much greater than c^* .

We have introduced a polydispersity parameter. This averaged structure factor is a simplified point of view because this approximation just leads to a smoothing of the structure peaks but does not influence the value of the structure factor for small q values. The exact structure factor for polydisperse systems, however, describes the behaviour at low q values precisely and therefore $S(0)$ is no longer directly related to the isothermal osmotic compressibility. For reasons of simplification we have restricted ourselves in this contribution to the approximation of the averaged structure factor. A detailed discussion of the effective structure factor for polydisperse systems and a comparison with the decoupling approximation is given elsewhere (Weyerich, Brunner-Popela & Glatter, 1997).

One of the most important points concerns the minimum search. The algorithm is based on the derivatives of the structure factor with respect to its parameters. We just need starting values for the different parameters of the structure factor, and in general it may be expected that such first estimates are known, *i.e.* the starting value for the radius can be checked by a first evaluation of the scattering data neglecting the interparticle contributions. Although there will appear oscillations in $p(r)$ at large r values, the maximum dimension of the particle can be roughly estimated (Glatter, 1979). With regard to the volume fraction it is not too difficult either, as the sample composition is known in most cases. The least-known parameter concerns the polydispersity. We have recognized that the minimum search works in a different way when we start near or far away from the global minimum. Therefore we are aware of the necessity to improve this algorithm and we are currently testing different algorithms.

In this contribution, we have only shown results from simulations as the application to real data would need a careful description of the systems under investigation, and the detailed discussion of the results would completely overload this contribution. For several different examples we were able to show that we can recover simultaneously the form factor and the structure factor. In these simulations we used the same models for the structure factor for the simulation and the evaluation step. We had 'only' to recover the correct parameters, *i.e.* to find good parameters for $S(q)$ and to recover $P(q)$ and $p(r)$ for different structures.

Here we want to stress the point that we were able to recover homogeneous and inhomogeneous structures (spherical and nonspherical) without assuming any model. The widely used model-fitting procedure needs such assumptions on the shape and structure. If these assumptions are correct one has to optimize only a few model parameters, and these results might be better if all assumptions apply. We believe that such model calculations might be helpful as a final step after applying our

technique, *i.e.* with the help of $P(q)$ and $p(r)$ determined with our techniques it will be much easier to find models that are justified by the data and not only by the ideas of the experimenter and his knowledge of the system.

We are currently applying this technique to SANS data from nonionic microemulsions with different compositions and volume fractions up to 30% and more. In this test, we also compare different models for the representation of the polydispersity for the structure factor. A corresponding paper is in preparation (Brunner-Popela, Glatter & Strey, 1997).

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